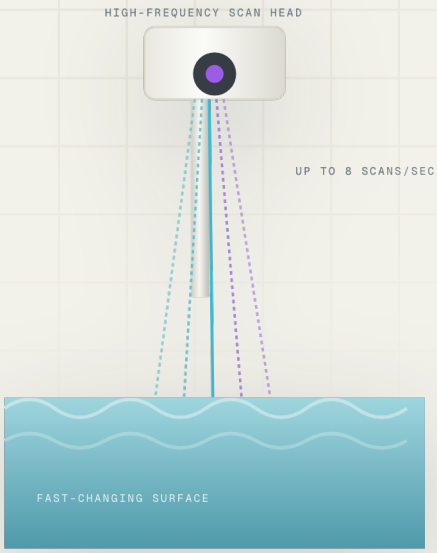
VEPOLINK
MONITORING PURITY

PROCESS INSTRUMENTATION × WATER LEVEL

HIGH-FREQUENCY RADAR FOR FAST-CHANGING WATER

RadarSense HF *Level Sensor*

The same 80 GHz FMCW radar platform, scanning up to eight times a second — built to keep pace with flash floods, surge, and water levels that change faster than a standard sampling rate can follow.



8 Hz
MAXIMUM SAMPLING FREQUENCY

±2 mm
LEVEL ACCURACY, 0-30M

100 ms
DURATION PER SCAN

8HZ SAMPLING FAST-CHANGING LEVELS | **SMART SENSORS** TILT & HUMIDITY | **VEPOLINK** FIELD-PROVEN BUILD

8Hz
HIGH-FREQUENCY SAMPLING OPTION

100ms
DURATION PER SCAN

1–60sec
ADJUSTABLE AVERAGING WINDOW

±2mm
ACCURACY, 0-30M RANGE

01 OVERVIEW

For water *that won't sit still.*

Built on the same 80 GHz FMCW radar platform as the standard RadarSense sensor, the HF variant adds a high-frequency sampling option — one, two, four or eight scans per second, each lasting 100 milliseconds — for sites where water level changes faster than routine monitoring can track.

Each measurement interval is the arithmetic mean of several scans over an adjustable averaging time of 1 to 60 seconds, balancing responsiveness against noise exactly as the site demands.

REPORTED CHANNELS

- WATER LEVEL / DEPTH TO WATER
- INTERNAL RELATIVE HUMIDITY
- SENSOR ORIENTATION (TILT)
- SDI-12 OUTPUT
- RS-485, MODBUS RTU
- INTERNAL DATA LOGGER

· PRODUCT FEATURES

Up to 8 Scans per Second

A high-frequency sampling option captures fast-changing levels that a once-per-second reading would smooth over or miss entirely.

Configurable Averaging

An adjustable 1–60 second averaging window (factory set to 1 second) tunes the balance between responsiveness and smoothing.

Same Smart Health Sensors

Built-in X/Y inclination and internal humidity monitoring carry over from the standard platform, unchanged.

Flood-Proof Housing

IP67-rated, with the lower housing tolerating full submersion for up to one week at a 1 m water column.

02 MEASUREMENT PRINCIPLE

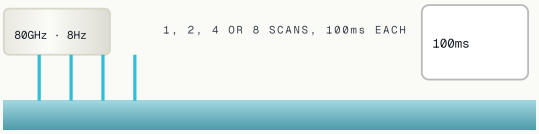
The same radar, *scanning faster.*

FMCW radar in the 77–81 GHz band provides the ranging; what the HF variant changes is how often it looks. Instead of one scan per second, it can take one, two, four or eight 100-millisecond scans and average them over a configurable window.

FMCW

High-Frequency Scanning

80 GHZ · 1, 2, 4 OR 8 SCANS/SEC



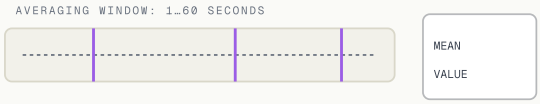
Each scan takes 100 milliseconds; running several per second captures detail that a single annual-average sampling rate would lose.

`100ms per scan`

PROCESSING

Adjustable Averaging

1–60 SEC WINDOW · FACTORY 1 SEC



Each measurement interval is the arithmetic mean of several scans over the configured averaging time, trading noise smoothing against responsiveness.

`Tune to the site`

SMART SENSING

Self-Monitoring Built In

X/Y INCLINATION · INTERNAL HUMIDITY



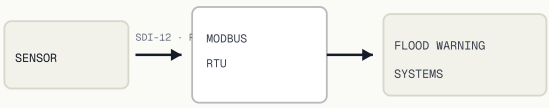
Internal inclination and humidity sensors report device health alongside the level reading, unchanged from the standard platform.

`Device health, remotely`

OUTPUT

Transmission & Reporting

SDI-12 · RS-485 (MODBUS RTU)



High-frequency readings flow into flood warning and SCADA systems over the same standard SDI-12 and Modbus RTU interfaces.

`Standard protocols, no extras`

03 CAPTURING FAST WATER

When one reading a second *isn't enough.*

Flash floods, gate releases and tidal surges can move a water surface meaningfully within a second or two. A sensor sampling once a second can miss the true peak, or report a level that's already out of date by the time it's logged.

SCENARIO 01

Flash floods

A rapidly rising stage during intense rainfall needs a sampling rate that tracks the rise, not just the eventual peak.

SCENARIO 02

Gate & sluice operation

Opening a dam gate or sluice can shift downstream level within seconds — fast sampling captures the transient, not just the new steady state.

SCENARIO 03

Tidal bores & surge

Storm surge and tidal bore fronts arrive abruptly; higher sampling resolves the leading edge rather than averaging through it.

TRADE-OFF

Averaging window

A shorter averaging window favours responsiveness; a longer one favours noise smoothing — both are configurable per site.

CHOOSING A SAMPLING RATE

Start from the fastest realistic rate of change at the site and set the scan count and averaging window so the reading keeps pace without amplifying surface noise from wind or turbulence.

DATA VOLUME

Higher sampling naturally produces more readings to log and transmit — plan telemetry bandwidth and logger memory accordingly for sustained 8 Hz operation.

04 SPECIFICATIONS

Technical *specifications.*

MEASUREMENT

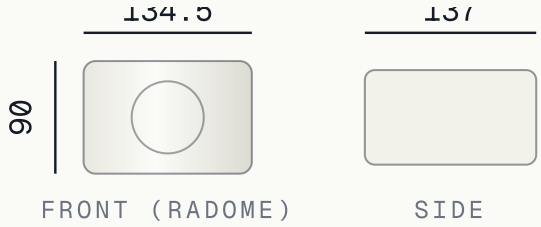
Parameter	Water level / depth to water
Working Principle	FMCW radar, 80 GHz (W-band)
Measuring Range	0 - 30 m, distance to water
Level Accuracy	± 2 mm
Resolution	0.001 m · 0.1 cm · 1 mm
Beam Angle	8°, no blanking distance
Sampling Option	1, 2, 4 or 8 scans/sec, 100 ms each
Averaging Time	1 - 60 sec, adjustable (factory: 1 sec)
Additional Parameters	Internal humidity · sensor orientation (X/Y)

SYSTEM · BUILD · POWER

Interface	SDI-12 & RS-485 (Modbus RTU)
Internal Data Logger	Yes
Power Supply	5.5 - 28.8 V DC, typ. 12/24 V DC
Power Consumption, Active	4 mA
Power Consumption, Idle	250 µA
Housing Material	Aluminium (AlMgSi1); ASA / PTFE radome
Mounting Material	Stainless steel 1.4301 (V2A)
Cable	PUR jacket, M9 connector
Protection Class	IP 67 – floods without damage, 1 week / 1 m water column
Dimensions (H×W×D)	90 × 134.5 × 137 mm

SENSOR HEAD DIMENSIONS

INDICATIVE FRONT & SIDE · MM



05 INSTALLATION & SITING

Mounted to *keep up with the site.*

> Mounting follows the same practice as the standard RadarSense platform — above the maximum expected water level, aimed straight down, beam cone clear of obstructions. The difference is in configuration: set the scan count and averaging window to match **how fast this specific site actually moves.**

01 · MOUNTING POINT

Bridge, pier or arm

Mount above the expected maximum water level, aimed straight down at the measurement point — no blanking distance to allow for.

02 · OBSTRUCTION CHECK

Low water inspection

Check for submerged obstructions such as rocks or piers at the lowest anticipated water level, before they are covered.

03 · SAMPLING RATE

Match the site's dynamics

Select 1, 2, 4 or 8 scans per second based on how fast the level genuinely changes at this location.

04 · AVERAGING WINDOW

Balance noise vs. speed

Set the 1–60 second averaging window — shorter for surge events, longer where surface noise needs smoothing.

05 · INTERFACE WIRING

SDI-12 or Modbus RTU

Wire into the site's datalogger or flood warning platform over SDI-12 or RS-485.

06 · TELEMETRY CHECK

Bandwidth for high-rate data

Confirm the telemetry link and logger memory can sustain the chosen sampling rate during a real event, not just in quiet conditions.

REMOTE HEALTH CHECKS

Tilt and humidity values remain available remotely alongside high-frequency level data, so sensor health can be confirmed without a dedicated field visit — important at sites that are hardest to reach exactly when they matter most.

06 APPLICATIONS

Where it *works.*

Sites where the water level can move meaningfully within a second or two, and a standard sampling rate would blur or miss what actually happened.

Flash Flood Warning	Dam & Gate Monitoring	Tidal Bores & Storm Surge
Hydraulic Structures	Stormwater Surge Basins	Urban Flood Warning

MIXED NETWORKS

Because it shares the same interfaces and mounting approach as the standard RadarSense sensor, the HF variant can be deployed at just the handful of sites that need fast sampling within an otherwise standard network.

EARLY WARNING SYSTEMS

Faster sampling shortens the time between a level change occurring and it reaching an alarm threshold in the warning system — a meaningful margin at sites with very short response windows.

07 FAQ

Frequently asked *questions.*

Q How is this different from the standard RadarSense sensor?

Same 80 GHz FMCW radar and accuracy, but with a configurable high-frequency sampling option of up to 8 scans per second instead of a fixed 2 Hz rate.

Q How long does each scan take?

100 milliseconds per scan, with 1, 2, 4 or 8 scans available per measurement interval.

Q What is the averaging window?

Adjustable from 1 to 60 seconds (1 to 480 samples), factory set to 1 second — shorter windows favour speed, longer windows favour smoothing.

Q How accurate is it?

±2 mm across the full 0–30 m measuring range, unchanged from the standard platform.

Q When should I choose this over the standard sensor?

Where water level can change meaningfully within a second or two — flash floods, gate operation, tidal bores and storm surge.

Q What outputs are available?

SDI-12 and RS-485 carrying Modbus RTU, the same as the standard platform.

Q Does it still have inclination and humidity monitoring?

Yes — both the X/Y inclination sensor and the internal humidity sensor carry over unchanged.

Q Does high-frequency sampling use more power?

Active current draw rises modestly with scan rate; idle current between measurement intervals remains at 250 µA.

Q Can it survive flooding?

Yes — IP67 rated, tolerating full submersion for up to one week at a 1 m water column.

Q Does high sampling need special telemetry?

Confirm bandwidth and logger memory can sustain the chosen rate during an actual event, not just at idle.

This brochure is general product information. Configuration, dimensions and performance figures are indicative and confirmed at order. For installation guidelines and full technical detail, refer to the user manual supplied with the product. Specifications may change without notice as the product is improved.

VEPOLINK
MONITORING PURITY

BUILT TO KEEP UP WHEN THE WATER DOESN'T SLOW DOWN

Fast water. *Faster readings.*

Tell us how quickly the level moves at your site, and we'll confirm the scan rate, averaging window and telemetry fit. Pick how you'd like to start.

01 **Talk to our engineers**
Speak directly to the team that sizes, specifies and commissions high-frequency radar level sensors. No call centre.
+91 99538 86178 →

02 **Request a site sizing** RECOMMENDED
Send your site's expected rate of change and telemetry platform — we'll confirm the sampling setup.
sales@vepolink.com →

03 **Book a demonstration**
See a running sensor at high sampling rate with our engineering team.
Request a demo →

WWW.VEPOLINK.COM · SALES@VEPOLINK.COM